

Overview of work for the past week



- (minor) bugfix.
- More MIMO.

Previously on MIMO

Log Code	Protocol Layer	MIMO Direction	Key MIMO Fields / Parameters
0xB887	5G NR MAC / PHY	Downlink (DL)	1 Num Layers 2 MCS CW0/CW1 3 TB Size
0xB883	5G NR MAC / PHY	Uplink (UL)	1 Num Layers (Rank) 2 Antenna Ports 3 SRS (Sounding Reference Signal)
0xB881	5G NR MAC / UCI	DL Feedback (in UL)	1 Num RI (Rank Indicator) 2 RI Value 3 CQI (Channel Quality Index) 4 New Tx Bytes (per RI)

Some confusion

- There are no 5G MIMO specification for the device on the official webpage.
- I asked Gemini about this and it said the hardware for NR(5G) and LTE(4G) must be same so no issue.
- I am bringing this up because in a previous meeting I had stated that MIMO is 2x2, but logically and empirically this does not seem to be the case.
- MIMO is now 4x4.

Wi-Fi

2x2 MIMO, Support 2.4G / 5G, Support WiFi 802.11 a/b/g/n/ac/ax

LTE/LTE-A

4x4 MIMO, 3CA, Support up to DL Cat 18(1100Mbps)/UL Cat 13(90Mbps)



more MIMO

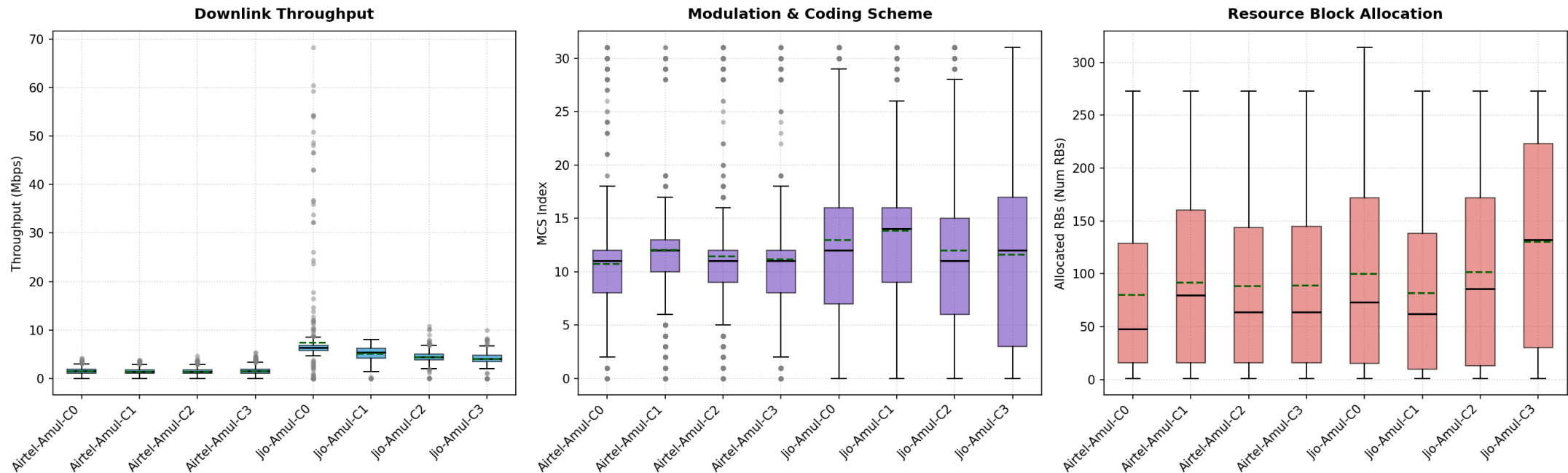
- MIMO (Multiple-Input Multiple-Output) uses multiple transmitters and receivers to multiply the capacity of a radio link.
 - What does **$n \times n$** MIMO mean? The numbers stand for the number of antennas communicating on each end of a wireless link.
 - **Transmit Antennas (Tx) $\times$ Receive Antennas (Rx). ON THE SAME LINK.**
 - Uplink 4x4 is rare.
 - Power is a major constraint and so is Space, since the transmitting antennas need to be physically separated so that the signals don't overlap destructively.
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Work on MIMO

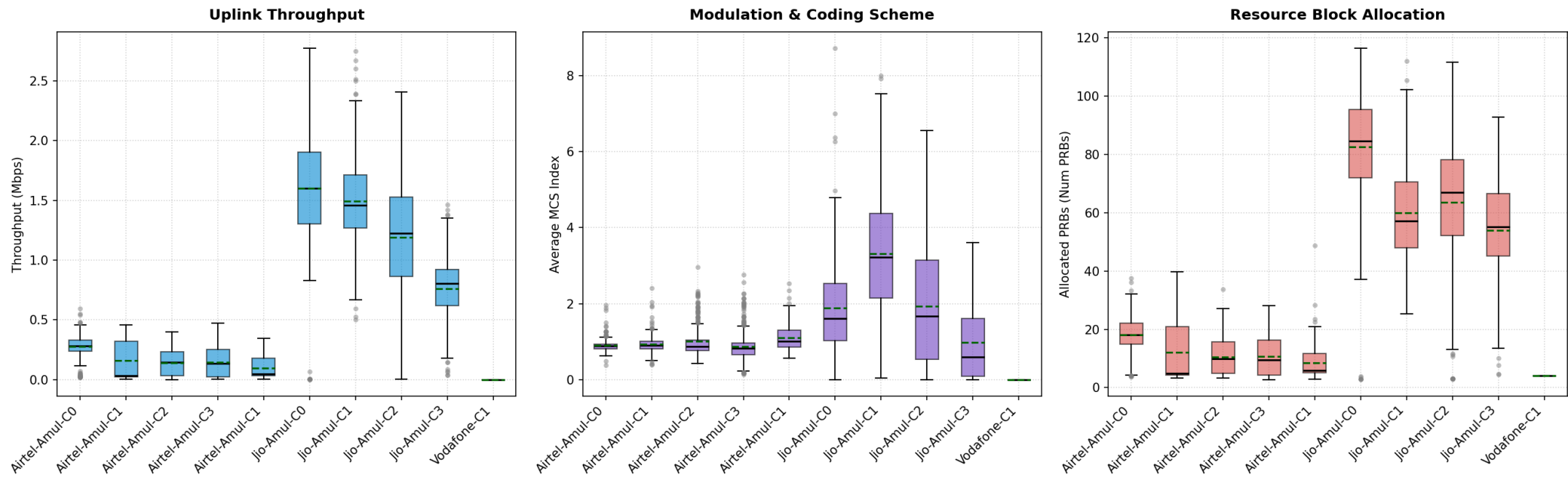
- Downlink: b887 parser updated to reflect number of layers.
- Uplink: b881 parser is giving weird numbers, maybe specifications changed a lot between version 2 and 3. So, I went with b883.

Total PUSCH Grant	Rank 2	Rank 1
1,213	1,211 (99.8%)	2 (0.2%)
1,690	1,689 (99.9%)	1 (0.1%)
1,439	1,439	0
1,225	1,225	0

Downlink Performance Metrics Distribution Across Runs



Uplink Performance Metrics Distribution Across Runs





What next

- Could explore b883 for uplink carrier aggregation.
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